

Assignment 4 - bonus point

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Exercise 2.1

In this section, I compare the sequential text training implemented in the basic model with the chunk shuffle training approach. 10% of the full dataset is reserved as the validation set. The smooth loss and validation loss for both training methods are shown in Fig.1, with the corresponding numerical values reported in Table 1. It can be observed that the chunk shuffle method consistently achieves a slightly lower validation loss than the basic sequential approach, indicating that there is a quantitative difference between the two methods.

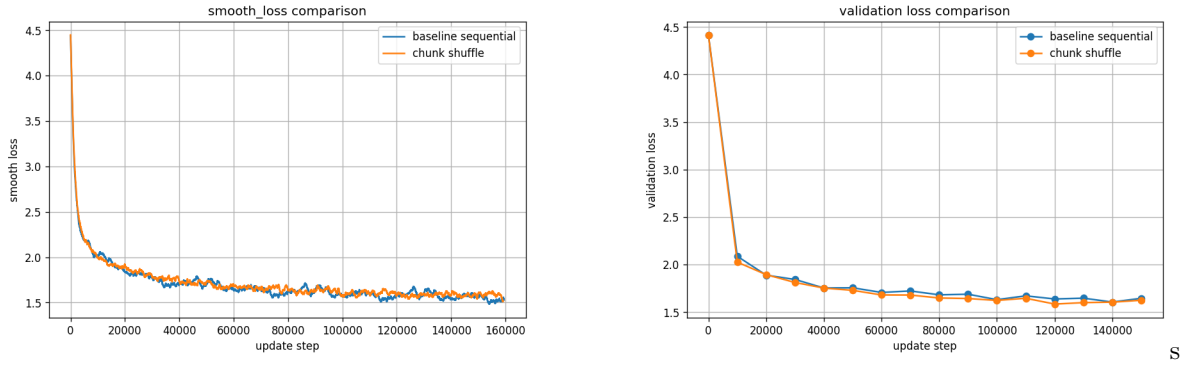


Figure 1: Performance of two different ways of training

Table 1: Validation Loss Comparison

Iteration	Baseline Val Loss	Chunk-Shuffle Val Loss
0	4.416128	4.416128
10,000	2.087915	2.026146
20,000	1.889157	1.894549
30,000	1.843799	1.811888
40,000	1.754214	1.753006
50,000	1.756004	1.729223
60,000	1.707096	1.680549
70,000	1.722769	1.679760
80,000	1.682177	1.648820
90,000	1.688518	1.643012
100,000	1.631079	1.623161
110,000	1.671575	1.645816
120,000	1.638319	1.585034
130,000	1.646925	1.599935
140,000	1.605974	1.604962
150,000	1.645669	1.624549

Exercise 2.2

In Bonus 2, I investigate the effect of using mini-batches larger than one by evaluating batch sizes $\{1, 8, 16, 32\}$. To ensure a fair comparison, all experiments share the same initialization seed and the same number of Adam update steps (100,000).

As shown in Fig. 2, larger batch sizes achieve a given smooth loss level with significantly fewer update steps. For example, the model with $bs=32$ (red curve) reaches a smooth loss of approximately 2.0 in around 4,000 updates, whereas the $bs=1$ setting (blue curve) requires about 10,000 updates. Moreover, at the same number of update steps, larger batch sizes consistently attain lower validation loss. These results indicate that increasing the batch size improves convergence efficiency with respect to update steps and leads to better-optimized models.

In addition, as demonstrated in Listing 1, qualitative text samples generated with $bs=16$ and $bs=32$ exhibit noticeably better coherence than those produced with $bs=1$, suggesting improved modeling of long-range dependencies.

Overall, these findings demonstrate that, in this setting, using mini-batches larger than one enhances both convergence speed and final model quality.

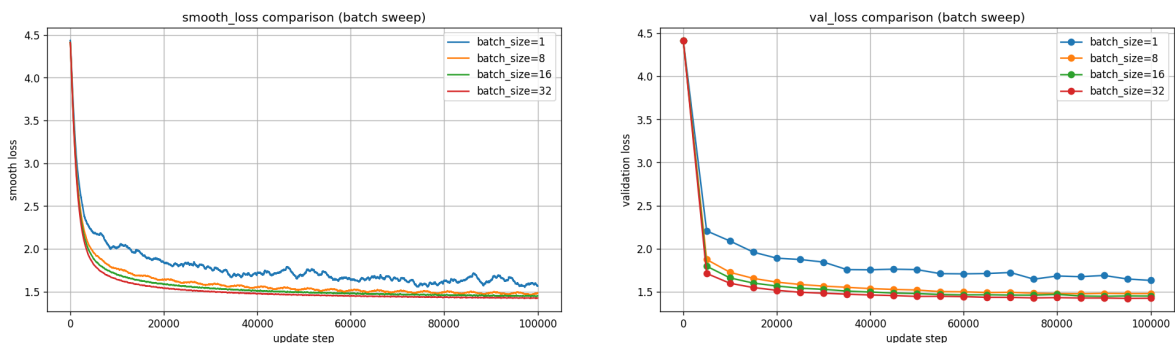


Figure 2: Performance of different batch sizes

Listing 1: Sampling of different batch_size

```
=== batch_size = 1  iter = 100000  smooth_loss = 1.5669 ===
them good if half onle. Wharray hard the harrs.
"But was going - sertong port half-knorge. But tell Rite," Harry sphat Ron,
    and he was don't he hat of himself the long orow a around he'd realing he

=== batch_size = 8  iter = 100000  smooth_loss = 1.4832 ===
sortifle's Snape. . . and there, back very downstafrest for brocarations, and
    of a knows of of theilly bewmet after the magick.
Fon's was excedped.
"Mudd's say up their blacker a coming in has been amo

=== batch_size = 16  iter = 100000  smooth_loss = 1.4499 ===
e's sire of the ophoud mets as the many of his walking.
"Malfoy , "Why sore porned out.. . me boico's getting about the pall, and is
    it raised another of eyes their time, looked around not ... he had

=== batch_size = 32  iter = 100000  smooth_loss = 1.4255 ===
ind. It seared the taked. "Well, immoned grown a sent, he foun ter your
    bright.
"No," Harry goddies dants that he was really corned was good and her laggidis
    them most of the OWhot with that and Ron
```

Exercise 2.3

In this section, I first train an RNN model and select the checkpoint with the lowest smooth loss as the best-performing RNN. Based on this fixed, pre-trained model, I then vary only the sampling strategy used to convert logits into a probability distribution during text generation, specifically comparing temperature sampling and nucleus sampling.

Temperature Sampling

When $T \in (0, 1]$, the distribution gets more peaky as T decreases. This is validated in Listing 2, where the temperature is set to $T \in [0.1, 0.5, 0.8]$. As T decreases, the output distribution becomes increasingly peaked:

- At $T = 0.1$, the highly peaked distribution leads to text degeneration, manifested as repetitive loops (e.g., “started”, “maze”, “the”), due to over-sampling of the highest-probability characters.
- As T increases to 0.8, the distribution becomes flatter, enabling more diverse vocabulary and more complex sentence structures. However, it also introduces non-existent words (e.g., “shadd”, “shouch”), indicating a trade-off in which increased diversity comes at the expense of linguistic accuracy.

Listing 2: Temperature Sampling

```
=====
[Temperature] low T   T = 0.1
=====
"The maze the see when Harry was started the started the started the spoke
  the maze.  He was said the maze of the maze the started the started the
  spoke the spoke the see when Harry was started the

=====
[Temperature] medium T   T = 0.5
=====
The pook Sare stared it face.  I wands unto the great still the started you
wand the askeel the wand.
"You a mare started the his propped and still you upen and what had ever down
  about it was rears

=====
[Temperature] high T   T = 0.8
=====
I felt," said Dumbledore said to his wand the wand come than watched even?"
Dumbledore shadd against shouch of momented him; have to catep . . . Weaclest
  pues were mean neared.
"We has can wizl he p

=====
```

Nucleus Sampling

The main advantage of nucleus sampling lies in its ability to adapt to the changing shape of the probability distribution at different stages of text generation. As observed in Listing 3:

- At $\theta = 0.2$, the generated text exhibits high repetitiveness, with frequent occurrences of words such as “maze”, “start”, and “spoke”. This is because the sampling pool is highly restricted, limiting the diversity of possible outputs.
- At $\theta = 0.5$, diversity increases, and the vocabulary becomes noticeably richer, with words such as “master”, “father”, and “shaking” appearing in the generated text.

- At $\theta = 0.9$, the model produces a wide range of proper nouns (e.g., “Dumbledore”, “Voldemort”) and more complex vocabulary (e.g., “cloughted”, “strolly”). However, this is accompanied by more spelling errors, such as “went” being generated as “men”, “around” as “arold”, and the creation of non-existent words like “starttrolley”.

Listing 3: Nucleus Sampling

```
[Nucleus] low theta  theta = 0.2
=====
"The paidn the spoke the maze.  He was started the spoke the maze.  He was
really the man was should the maze the spoke the see when he was said go
deep and show should the maze.  He was started the

=====
[Nucleus] medium theta  theta = 0.5
=====
"The started the shoke face what hands for the master strented the caught on
the father shaking shoulder hands of perslest which the maze of come the
fore to the parter that wand he pulled the groun

=====
[Nucleus] high theta  theta = 0.9
=====
Then his cloughted more a keace was stopper where a ment up and say, and
pook sens arold ears head," said Dumbledore again.  They had of Voldemort
could mean of the standed his starttrolley.  "My from

=====
```